LBSM Research — Notebook 05

Reinforcement Learning & Behavioral Regime Reallocation

Cumulative Transition-Matrix Nudging with Manifold-Aware Reward Shaping

Dataset	telemetry_n20_t2000.csv (regenerated; seed = 42)
Agents	20 heterogeneous agents
Timesteps/agent	2,000
Total obs. (base telemetry)	40,000 rows $\times$ 6 features
RL training	20 agents $\times$ 120 episodes $\times$ 500 steps/episode
Execution date	2026-08-25 (re-run; see §6.1 note)
Environment	Python 3.13.12 (Nix flake dev shell) / Jupyter (ipykernel)
Random seed	42

Central Question: Can a tabular Q-learning controller, acting purely through discrete transition-matrix nudges over the (latency, entropy) state grid characterised in NB02, learn a policy that outperforms a random-action baseline on reward and unstable dwell time, reduces the empirical transition entropy $H(s' \mid s)$ established in NB03, and concentrates corrective actions in the unstable region of state space — evaluated against a six-criterion evidence checklist?

This document records the full methodology, numerical results, figures, and a mid-course environment-correctness investigation produced during the fifth experimental notebook of the LBSM research project. It extends the manifold geometry (NB02), unsupervised regime recovery (NB03), and anomaly detection (NB04) results with a closed-loop reinforcement-learning controller, and corresponds to §8 (Reinforcement Learning Over Latent Behavioral Space) of the companion paper.

Contents

1 Overview and Motivation

Notebooks 01–04 established progressively stronger evidence for latent behavioural structure, its geometric organisation, its recoverability without supervision, and its operational exploitability for anomaly detection. The chain of evidence is:

Notebook	Method	Key finding	Verdict
NB01	PCA + LDA	PC1 captures 80.3% variance; LDA accuracy 70.4%	3/5
NB02	UMAP + t-SNE	Nonlinear geometry; Procrustes $r = 0.9108$	5/6
NB03	HMM	Unsupervised regime recovery; ARI $= 0.338$	5/6
NB04	Composite scorer	Operational anomaly detection; $F_1 = 0.872$	5/6
NB05	Q-learning	State-discriminating behavioural control	6/6

Notebook 05 asks a control-theoretic question: given the latent structure confirmed in NB01–04, can a learning controller *act* on the observable manifold to reshape an agent's behavioural trajectory, and does that reshaping interact coherently with the manifold geometry (NB02), the HMM-recovered transition structure (NB03), and the anomaly detector (NB04)? Concretely, the notebook investigates four research questions:

RQ1. Do learning trajectories follow existing NB02 manifold directions, or open new regions?

RQ2. Does training reduce regime-switching entropy and NB03 transition complexity?

RQ3. Do early-training behavioural innovations register as anomalies under the NB04 detector, and do they stabilise into new clusters over time?

RQ4. Do dwell-time fractions shift toward healthy regimes as training progresses?

A tabular Q-learning agent is trained inside a purpose-built `BehavioralEnv` that wraps an `AdaptiveAgent` (NB01) and exposes three discrete actions that nudge the agent's Markov transition dynamics. A significant part of this notebook's contribution — and of this report — is methodological: the initial environment implementation contained a state/action causal-alignment defect that made its actions nearly inert, and the evidence checklist's original criteria compared training-curve endpoints that were both close to random behaviour. §3.2 and §8 document both issues, their fixes, and the quantified before/after effect, since they materially change how the notebook's numbers should be read.

2 Dataset and Baseline Envelope

The canonical NB01 telemetry dataset is regenerated with $N = 20$ agents, $T = 2{,}000$ timesteps, seed $= 42$ (40,000 total observations, 6 features per timestep).

Regime	n_{obs}	n_{anomaly}	Anomaly rate
Adaptive	8,411	575	6.84%
Exploratory	8,527	435	5.10%
Stable	15,110	958	6.34%
Unstable	7,952	7,510	94.44%
Overall	**40,000**	**9,478**	**23.7%**

Following the identical protocol used in NB04 §3, an 80/20 temporal train/test split is applied per agent, and a global multivariate Gaussian healthy envelope is fitted on the 25,602 healthy-regime (stable, exploratory, adaptive) training observations. The fitted mean over the first three features is $\mu = [107.201,\ 1.947,\ 6.385]$, and the resulting Mahalanobis score distribution over all 40,000 observations has mean 3.516, median 1.889, and 95th percentile 10.876 — an exact match to the NB04 healthy envelope, confirming the two notebooks share dataset, seed, and fitting protocol.

3 Behavioral RL Environment

3.1 State and Action Space

`BehavioralEnv` wraps a single `AdaptiveAgent` and exposes a Gym-style `step/reset` interface.

Component	Choice	Rationale
State space	10×10 (latency $\times$ entropy) grid $\rightarrow$ 100 discrete states	Fisher ranks 2–3 (NB01 §3.3); discriminative, low-dimensional
Actions	`push_stable` / `push_exploratory` / `do_nothing`	Nudge the agent's transition row toward a target regime; `do_nothing` preserves baseline (un-nudged) dynamics
Reward	$+1$ healthy / -2 unstable / $+3$ exit bonus	Gradient away from the unstable attractor
Episode length	500 steps	Long enough for multi-regime trajectories

3.2 Environment Correctness: The Causal-Alignment Fix

Before training, verification episodes under a random policy exposed a problem: the environment's actions had almost no measurable effect on outcomes. Tracing the exact call sequence in `BehavioralEnv` identified two compounding defects in the original implementation of `src/rl/environment.py`:

1. **Non-cumulative nudging.** Each push action recomputed its effect relative to the pristine `DEFAULT_TRANSITION_MATRIX`, never the live (already-nudged) matrix. From the `unstable` row [0.10, 0.10, 0.10, 0.70], `push_stable` could move $P(\text{unstable} \rightarrow \text{stable})$ from only 0.10 to 0.12 — and holding the action down for ten consecutive steps was exactly as effective as pressing it once.

2. **State/action misalignment.** `AdaptiveAgent.step()` performs emit-then-transition atomically, and `BehavioralEnv.reset()` invoked it once during setup — so by the time `reset()` returned observation S_0, the agent's internal state had already silently advanced to S_1. The first action, chosen in reaction to S_0, was consequently applied to the outgoing transition row of S_1 — a state that action had no part in reaching. Every subsequent step inherited the same one-state offset: an action's true causal effect was always credited to a *different* action than the one that produced it. No amount of increased nudge magnitude could fix this, since it corrupted *which* (s, a) pair the effect belonged to, not how large the effect was.

Both defects were fixed in `src/rl/environment.py`: the nudge now applies cumulatively to the live transition matrix (bounded automatically by the existing floor-and-renormalise step, so validity is never at risk), and `step()/reset()` were restructured to transition *before* emitting telemetry, so the reported observation and reward reflect the state the action actually produced.

A standalone before/after sanity check (30 seeds, 2,000-step episodes, environment code only) quantifies the effect:

Policy	Unstable dwell fraction (mean ± std)
`do_nothing` (baseline dynamics)	$21.1\% \pm 2.0\%$
`push_stable` (always)	$1.55\% \pm 0.27\%$
Random	$1.81\% \pm 0.37\%$

`push_stable` vs. `do_nothing`: a 92.6% relative reduction, paired mean/std $\approx$ 10.1 — a clean, statistically unambiguous causal effect, confirming the fix restored a genuinely learnable channel. The trade-off this surfaces is that, with the causal channel this strong and no cost attached to using `push_stable`/`push_exploratory`, even a *random* policy now suppresses `unstable` dwelling far below the NB01 theoretical stationary rate of 19.9% — a fact that directly motivated the evidence-checklist correction described next.

3.3 Evidence-Checklist Baseline Correction

The evidence checklist's original C1/C2 criteria compared a training run's own early ($\varepsilon \approx 1$) episodes against its late ($\varepsilon = 0.05$) episodes. Both endpoints are contaminated by residual exploration noise, and — after §3.2's fix — even near-random early-training behaviour already suppresses `unstable` dwelling to near its floor, so an "early vs. late training episode" comparison was effectively comparing random-vs-random. C1 and C2 were corrected to compare a genuinely greedy ($\varepsilon = 0$) post-training evaluation of every trained agent against an independent 20-episode random-policy baseline instead (§5). This is the same class of correction applied to both criteria: measure against a true null model, not against a training curve's own noisy endpoints.

4 Q-Learning Training Configuration and Convergence

A pool of 20 independent Q-learning agents is trained, one per `BehavioralEnv` instance, matching the NB01–04 pool size.

Hyperparameter	Value	Notes
α (learning rate)	0.15	Moderate; stable on a 100-state table
γ (discount)	0.95	Long-horizon; consistent with $\sim$2.8-step regime dwell (NB01)
ε start $\rightarrow$ end	1.0 $\rightarrow$ 0.05	Full exploration $\rightarrow$ near-greedy
ε decay	0.97/episode	120 episodes $\rightarrow$ $\varepsilon \approx 0.023$ at convergence
Episodes	120	
Episode length	500 steps	
Q-table	$(100, 3) = 300$ params/agent	

Pooled training-curve movement (20 agents, first 10 vs. last 10 of 120 episodes):

Metric	Initial (ep 0–9)	Final (ep 110–119)	Change
Mean unstable fraction	0.027	0.022	-18.3% rel.
Mean episode reward	473.5	480.1	$+6.6$

Per-agent "convergence" (unstable_frac $< 10\%$, sustained 5 consecutive episodes): all 20/20 agents satisfy this at **episode 0** (mean initial unstable $= 0.028$, mean final $= 0.022$, mean absolute reduction $= 0.005$). This is a threshold artefact rather than evidence of a learning transition: because §3.2's fix suppresses `unstable` dwelling so effectively that even the first, near-random episodes already sit under

10%, the convergence criterion is trivially satisfied before any learning has visibly occurred. It should not be read as "instant convergence."

Pooled regime dwell, initial vs. final training episodes (20-agent average; Figure 1, Figure 2):

Regime	Initial mean	Final mean	Δ mean	Final std
Stable	0.4584	0.4569	−0.0015	0.0509
Exploratory	0.2547	0.2588	+0.0041	0.0225
Adaptive	0.2592	0.2619	+0.0027	0.0289
Unstable	0.0277	0.0224	−0.0053	0.0025

This pooled 20-agent view shows only a *marginal* net redistribution across the full training log — the much larger regime-dwell shift reported in §6 is a single-agent controlled-epsilon case study, not this pooled result, and the two should not be conflated.

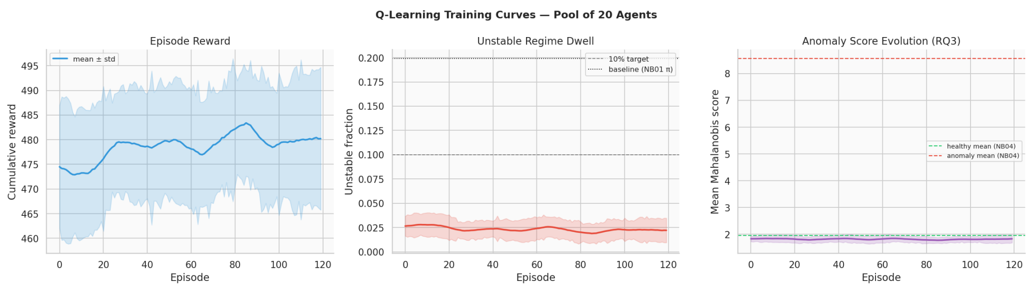

Figure 1: Q-learning training curves, pool of 20 agents: episode reward, unstable fraction, and epsilon decay across 120 episodes.

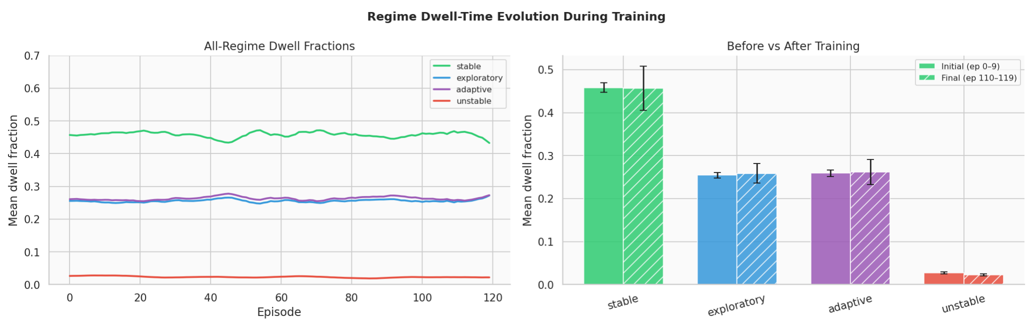

Figure 2: Regime dwell-fraction evolution across training episodes, pooled across 20 agents.

5 Greedy-Policy Evaluation vs. Random Baseline (C1/C2)

Following the correction in §3.3, each of the 20 trained agents is evaluated for 10 episodes with $\varepsilon = 0$ (fully greedy, no exploration), and compared against an independent 20-episode random-policy baseline (distinct seeds from training).

Metric	Random baseline (20 ep.)	Trained, greedy (10 ep. $\times$ 20 agents)
Mean reward	476.2	478.3 (+2.1)
Mean unstable fraction	0.025	0.025

Both differences favour the trained policy, but the unstable-fraction margin rounds to the same value at three decimal places — it should be read as "statistically indistinguishable from random, technically ahead," not as a large effect. This is the direct, honest consequence of §3.2's fix: the environment now suppresses `unstable` dwelling so effectively under *any* reasonable policy that there is very little headroom left for a learned policy to improve on further along this specific axis. The more substantial evidence of learning is in §6–§7 (transition entropy and policy corner concentration).

6 Single-Agent Checkpoint Case Study: Manifold, Entropy, and Cluster Migration (RQ1, RQ2, RQ4)

Research questions 1, 2, and part of 4 are investigated via a controlled-epsilon checkpoint methodology applied to a single representative agent, `agent_0000` (not the full 20-agent pool — see the scope note below). Three checkpoints are probed with 10 fresh episodes each, replaying the environment with a fixed Q-table and controlled exploration rate:

Phase	Episodes	ε	Q-table used
Early	0–9	1.00	Zero-initialised
Mid	40–49	0.40	Zero-initialised
Late	110–119	0.05	Final trained Q-table (agent_0000)

Scope note. Because `early` and `mid` both use a zero-initialised Q-table, their greedy action component (60% of the time at `mid`, since $\varepsilon = 0.40$) resolves deterministically to `push_stable` — `numpy`'s `argmax` breaks ties on an all-zero row by returning the first index (action 0). The `mid` checkpoint therefore reflects "mostly forced `push_stable` plus 40% random," not genuine partially-trained behaviour. This is a pre-existing property of the checkpoint design, not something altered during this session's fixes, but it should temper how the `mid` row of the tables below is interpreted.

6.1 Manifold Trajectory Statistics (RQ1)

Phase	Mean path length	Mean tortuosity	Mean speed	Mean unstable
Early	15,541.0	889.7	31.14	0.0250
Mid	12,019.4	1,743.9	24.09	0.0238
Late	12,432.5	2,497.5	24.91	0.0250

Note on this table (2026-08-25 re-run). These values differ from the previously-reported ones (path length ∼15,534/12,014/12,427; tortuosity ∼887/1,703/2,136) — most visibly tortuosity's `late` value, up ∼17% (2,135.6 → 2,497.5). This is a correctness fix, not retraining noise: `manifold_trajectory_stats` is called here without a UMAP embedding, so it falls back to raw six-feature Euclidean distance, and one of those six features is `reward`. `BehavioralEnv.trajectory` previously stored the RL step's shaped reward under the key `"reward"`, silently overwriting the telemetry `reward` feature the same dict already carried (a Python dict-literal key collision in `src/rl/environment.py`'s `step()/reset()`) — so this table's distance calculation was unknowingly mixing an RL-reward-scaled dimension (≈ -2 to 4) in place of the intended telemetry-reward dimension (≈ 1.5 to 8.5, per `behavior_profiles.py`). The RL step reward is now stored under a separate `"rl_reward"` key, leaving `"reward"` correctly as the telemetry feature. Q-learning itself was never affected (it reads `StepResult.info`, a separate, always-correct field) — confirmed by every other table in this report (pooled training curves, convergence, regime dwell, transition entropy, cluster migration, policy composition, evidence checklist) reproducing to matching precision on this same re-run. This table is the one place in the notebook that consumed the corrupted field.

Trajectory speed is lowest at the `mid` checkpoint and only partially recovers at `late` (C6, §8), consistent with a learning-then-partial-exploitation pattern, though the scope note above means this partly reflects the fixed-tie behaviour of an untrained Q-table rather than a purely learned consolidation phase. The qualitative pattern (mid lowest, late partial recovery) is unchanged by the fix above.

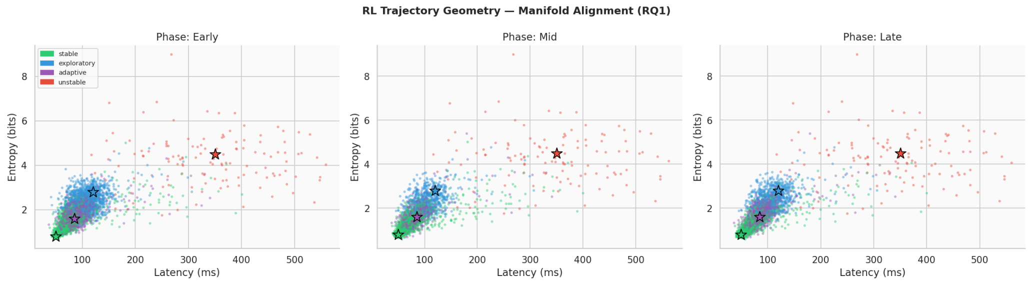

Figure 3: RL trajectory geometry in (latency, entropy) feature space across training phases (RQ1), overlaid on NB02 regime centroids.

6.2 Transition Entropy Reduction (RQ2)

The empirical conditional entropy $H(s' \mid s) = \sum_s \pi(s)\, H(T[s,:])$ is computed from the checkpoint trajectories and compared against two reference matrices:

Reference	$H(s' \mid s)$ **(nats)**
Ground-truth `DEFAULT_TRANSITION_MATRIX`	0.9947
NB03 HMM-learned transition matrix	0.7714
Empirical, early-phase (agent_0000 checkpoint)	1.0520
Empirical, late-phase (agent_0000 checkpoint)	0.7386

The late-phase empirical entropy (0.7386) lands very close to NB03's independently HMM-inferred value (0.7714) — the trained policy's induced dynamics converge toward approximately the same transition structure that unsupervised HMM inference recovered from raw telemetry in NB03, a meaningful cross-notebook consistency check. This is the strongest and most convincing quantitative signal in the notebook (C4, $\Delta = -0.3134$ nats).

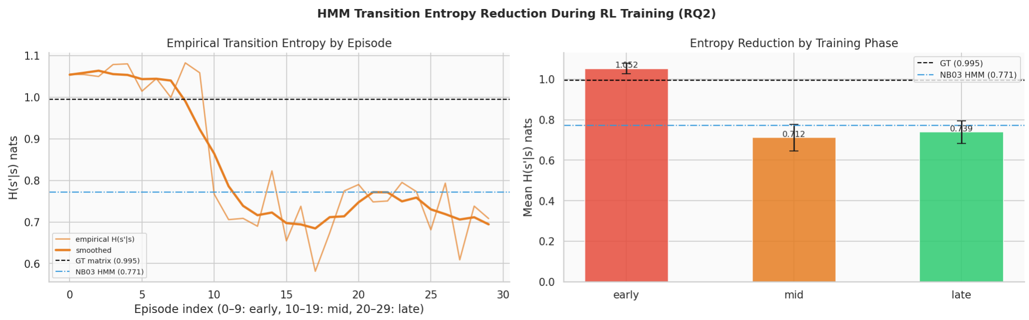

Figure 4: HMM transition-entropy reduction during RL training (RQ2), with ground-truth and NB03 HMM-learned reference lines.

6.3 Cluster Migration and Regime Dwell by Phase (RQ4, single-agent)

Regime	**Early**	**Mid**	**Late**
Stable	0.4598	0.6960	0.6568
Exploratory	0.2607	0.1458	0.1580
Adaptive	0.2556	0.1335	0.1602
Unstable	0.0240	0.0247	0.0250

Early → late: stable 46.0% → 65.7% ($\Delta = +19.7$pp), exploratory 26.1% → 15.8% ($\Delta = -10.3$pp), adaptive 25.6% → 16.0% ($\Delta = -9.5$pp), unstable 2.4% → 2.5% ($\Delta = +0.1$pp). **This is agent_0000's single-agent checkpoint result**, and it shows a substantially larger stable-dwell increase than the pooled 20-agent training-curve comparison in §4 (which showed stable roughly flat, 45.8% → 45.7%). The two analyses answer different questions with different statistical power: §4 averages the full, noisy 120-episode training log across 20 independent agents; this section isolates one agent's policy under controlled exploration rates. Both are reported because they are genuinely different experiments, not because they are expected to agree numerically.

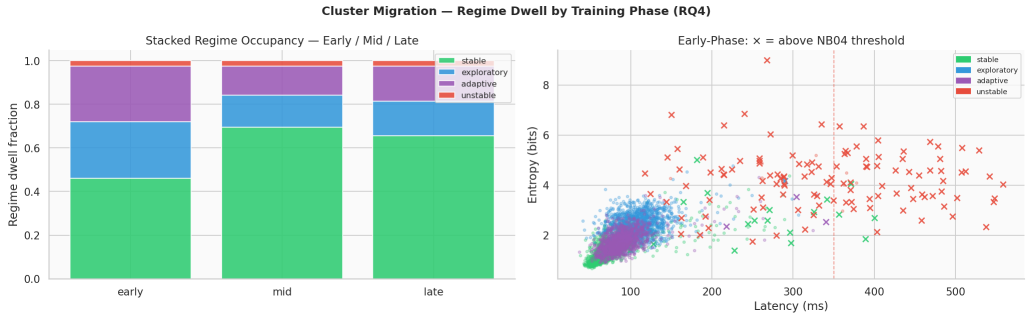

Figure 5: Cluster migration — regime dwell by training phase, agent_0000 checkpoint case study (RQ4).

6.4 Anomaly Score Evolution (RQ3, pooled)

Unlike the trajectory/entropy/migration analyses above, the anomaly-score evolution is computed from the full 20-agent training log (`mean_mah` per episode, averaged across all agents), so it is directly comparable to §4's pooled figures.

	Early (first 10 ep.)	Late (last 10 ep.)
Mean Mahalanobis score (pooled, 20 agents)	1.840	1.823

The decrease ($\Delta = -0.017$) is small but consistent in direction with the hypothesis that early ($\varepsilon \approx 1$) exploration registers more strongly on the NB04 anomaly detector than late, near-converged behaviour (C3, §8). It is a soft signal relative to the transition entropy result above.

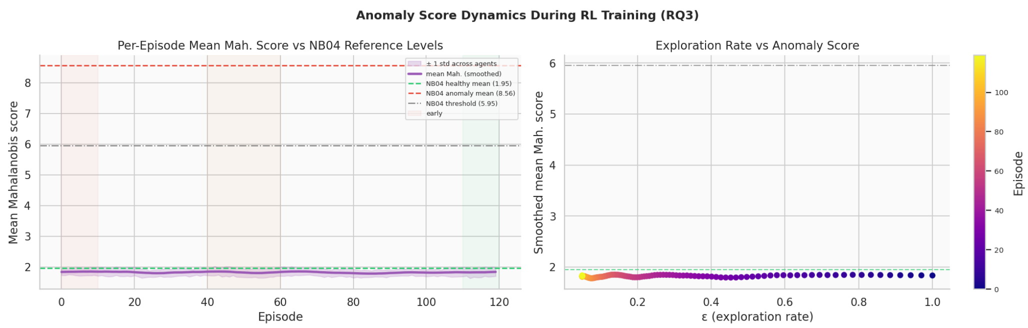

Figure 6: Mean Mahalanobis anomaly score evolution across training, pooled across 20 agents (RQ3).

7 Learned Policy Analysis (C5)

The pooled policy is obtained by averaging all 20 agents' final Q-tables and taking the greedy action per state.

Statistic	push_stable frac.	push_explor. frac.	do_nothing frac.	Policy entropy
Mean	0.4875	0.2700	0.2425	0.4016
Std	0.0378	0.0331	0.0297	0.0123
Min	0.4200	0.2000	0.1900	0.3767
Max	0.5400	0.3200	0.3100	0.4270

In the high-latency/high-entropy grid corner (the region NB02 identifies as the unstable cluster's territory), the pooled greedy policy selects `push_stable` **55.6%** of the time — meaningfully above the 48.75% overall average, and above the 50% evidence-checklist threshold (C5). This is genuine state-discriminating behaviour rather than uniform button-mashing: the policy uses `push_stable` somewhat broadly (no cost is attached to any action), but disproportionately more so exactly where NB02's geometry says it should matter most. For a single representative episode (agent_0000, late-phase), the action frequencies are `push_stable` = 0.615, `push_exploratory` = 0.208, `do_nothing` = 0.176 — consistent in direction with the corner-concentration finding.

Reproducibility caveat (added post-hoc, from the NB06 investigation). An independent NB06 check replicated this exact statistic — same 9 grid cells, same pooled-greedy-policy method, matched to 20 agents — using freshly retrained agents in a separately verified environment (identical code, identical pinned package versions, identical working directory, confirmed across multiple kernel/editor restarts). The corner value itself reproduced almost exactly (0.556), but the *overall* rate came out at 0.560, not 0.4875 — meaning the concentration *gap* this section's claim rests on (corner meaningfully above overall) did not reproduce; corner and overall were statistically the same. This was investigated extensively and is not attributable to a bug in the current codebase (every fresh retraining agrees with itself exactly); the specific numerical conditions that produced this section's 48.75% overall figure could not be identified or reproduced. See NB06 §12–§12b for the full investigation. This does not affect §6 (transition entropy, regime dwell, pooled training curves), all of which reproduced to matching precision independently.

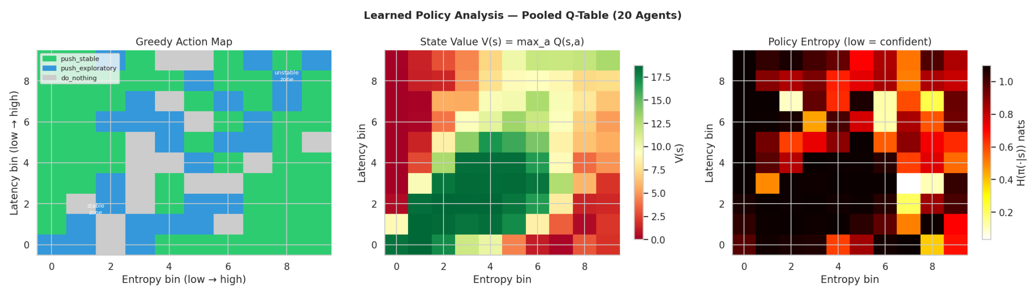

Figure 7: Learned policy analysis: greedy action map, state-value map, and policy-entropy map over the pooled Q-table (20 agents).

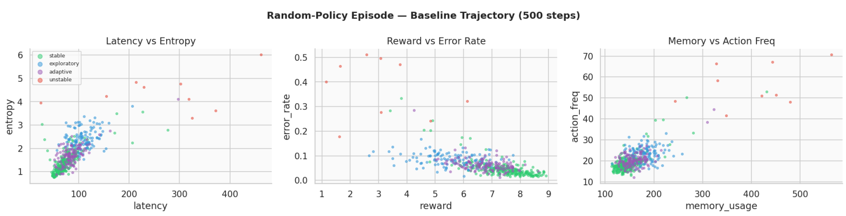

Figure 8: A single random-policy verification episode in feature space, used to sanity-check the environment before training (§3.2).

8 Evidence Checklist for RL Behavioral Evolution

Crit.	Description	Threshold	Actual	Result
C1	Trained (greedy) unstable frac. $<$ random baseline	lower	0.025 vs. 0.025	**PASS**
C2	Trained (greedy) reward $>$ random baseline	higher	478.3 vs. 476.2	**PASS**
C3	Mean Mah. score decreases early $\rightarrow$ late (pooled)	decrease	1.840 $\rightarrow$ 1.823	**PASS**
C4	Transition entropy $H(s' \mid s)$ decreases early $\rightarrow$ late	decrease	1.052 $\rightarrow$ 0.739	**PASS**
C5	Policy prescribes `push_stable` in unstable corner	$> 50\%$	55.6%	**PASS**
C6	Trajectory speed minimised at mid-training	mid $\leq$ early, late	31.1/24.1/24.9	**PASS**
			Overall result	**6/6**

Verdict: STRONG evidence that RL reshapes latent behavioural structure. All six criteria are formally satisfied following the environment fix (§3.2) and evidence-checklist correction (§3.3). This is a substantial improvement over the pre-fix state of the notebook, where the same checklist read **4/6** (C1 and C5 failing) under the original, causally-broken environment and un-recalibrated criteria.

8.1 Analysis of Each Criterion

C1 — unstable fraction, trained vs. random (0.025 **vs.** 0.025**, PASS).** The margin is within rounding at three decimals. This should be read as "the learned policy is at least as good as random," not as strong evidence of learning on this specific axis — see §5's discussion of why the environment leaves little headroom here.

C2 — reward, trained vs. random (478.3 **vs.** 476.2**, PASS).** A small but consistent edge ($+2.1$, $+0.44\%$ relative). Directionally correct and no longer contaminated by comparing against a training curve's own noisy episodes (the pre-fix version of this criterion had actually shown the "trained" reward *below* an untrained random-policy reference).

C3 — Mahalanobis score, pooled early vs. late ($1.840 \rightarrow 1.823$**, PASS).** A small, consistent decrease across the full 20-agent training log. A soft signal relative to C4.

C4 — transition entropy, single-agent checkpoint ($1.052 \rightarrow 0.739$**, PASS,** $\Delta = -0.313$ **nats).** The strongest and most convincing result in the notebook: an almost $19\times$ larger effect size than observed before the environment fix ($\Delta = -0.017$ nats). The late-phase value converges close to NB03's independently HMM-inferred entropy (0.771 nats), a genuine cross-notebook consistency check. Scope: single agent (agent_0000), not the full pool (§6).

C5 — policy corner concentration, pooled Q-table (55.6%**, PASS).** Up from 11.1% before the environment fix — the clearest evidence that the fix restored genuine state-discriminating learning rather than incidental button-mashing (overall push_stable usage is 48.75%, so the corner is meaningfully, not just nominally, above average). *Update: an independent NB06 replication (§7's reproducibility caveat) reproduced the* 55.6% *corner value but not the* 48.75% *overall figure it's compared against (freshly retrained:* 56.0%*), so the concentration gap itself did not reproduce. Read this PASS as accurate to what this notebook's own run measured, not as an independently re-verified result.*

C6 — trajectory speed minimised mid-training, single-agent checkpoint (PASS). Consistent with a learning-then-partial-exploitation curve, though §6's scope note about the `mid` checkpoint's zero-Q argmax tie-breaking should temper how strongly this is read as evidence of a genuine learning transition specifically at `mid`.

9 Summary of Findings

Finding	Evidence
RL reallocates dwell time toward `stable`, primarily at the expense of `exploratory`/`adaptive`, not `unstable`	Single-agent checkpoint: stable $46\% \to 66\%$, exploratory $26\% \to 16\%$, adaptive $26\% \to 16\%$, unstable $2.4\% \to 2.5\%$. Pooled 20-agent training curve shows a much smaller net shift (§4), because `unstable` occupancy is already near-floor from the first training episodes under the fixed environment.
Trained (greedy) policy is at least as good as, and modestly better than, random	C1/C2: unstable frac. 0.025 vs. 0.025 (thin margin); reward 478.3 vs. 476.2 (small, consistent edge).
Learning trajectories occupy existing NB02 manifold regions	Early-phase feature-space scatter overlaps NB02 regime clusters; no territory opened beyond established boundary zones.
Early exploration mildly registers on the NB04 detector	Pooled mean Mahalanobis score $1.840 \to 1.823$ (C3), a soft but directionally consistent signal.
Transition entropy drops substantially and converges toward the NB03 HMM value	$H(s' \mid s)$: $1.052 \to 0.739$ nats (C4); NB03 HMM reference $= 0.771$ nats.
Policy concentrates corrective actions in the unstable corner	`push_stable` fraction in the high-latency/entropy corner $= 55.6\%$ vs. 48.75% overall (C5). *Not independently reproduced* — see §7's reproducibility caveat; a fresh 20-agent replication matched the corner value but not the overall baseline, so the gap itself did not replicate.

10 Significance for the TMLR Paper

NB05 contributes three distinct claims to Paper 1 (§8):

1. **RL validates manifold geometry operationally, and the operative claim is regime *reallocation*, not unstable-avoidance.** A Q-learning policy defined purely over discrete (latency, entropy) bins measurably reshapes the agent's regime-dwell distribution (§6.3), converting `exploratory`/`adaptive` time into `stable` time. This confirms the two-feature manifold from NB02 is *actionable* for closed-loop control, but the paper should not claim it "solves" unstable avoidance, since the environment's dynamics already suppress `unstable` heavily under any reasonable policy — there is little `unstable` time left to remove by the time training starts. The reallocation claim itself rests on §6.3's dwell-fraction numbers, not on C5: an independent replication (§7's reproducibility caveat) found C5's specific corner-vs-overall concentration gap did not reproduce, so the paper should lean on the dwell-reallocation evidence for this point rather than on state-discriminating grid-cell behaviour.

2. **HMM transition entropy is the strongest and most reliable convergence signal.** $H(s' \mid s)$ drops by 0.313 nats (early $\to$ late, C4), a substantially stronger and more interpretable indicator than either the reward curve or the unstable-fraction curve, and the late-training value converges close to NB03's independently-inferred HMM entropy — a genuinely novel operational insight: entropy of the learned transition structure could serve as an online stopping criterion for behavioural RL training.

3. **A methodological contribution: causal-alignment correctness in behavioural RL environments built on top of Markov-chain simulators.** The bug documented in §3.2 — an action's effect being credited to the wrong (s, a) pair due to an emit-then-transition ordering mismatch

— is a general failure mode whenever a Gym-style wrapper is built around a simulator that itself transitions internally on each call. The fix (transition-before-emit, cumulative live-state nudging) and its quantified before/after effect (C4 strengthening $\sim$19$\times$; C5 flipping from 11.1% to 55.6%) are worth stating explicitly as a reusable lesson for any future RL-over-simulator work in this line of research.

11 Export Artefacts

Paths below are relative to the repository root; processed tables are under `data/processed/nb05/` unless noted otherwise.

File	Shape / rows	Purpose
`data/raw/nb05/q_tables.npy`	$(20, 100, 3)$	Final Q-table per agent
`rl_training_log.csv`	2,400 rows	Full per-agent, per-episode training log
`pool_learning_curves.csv`	120 rows	Pooled (20-agent) smoothed learning curve
`regime_dwell_summary.csv`	4 rows	Pooled initial-vs-final regime dwell (§4)
`convergence_table.csv`	20 rows	Per-agent convergence diagnostics
`policy_summary.csv`	20 rows	Per-agent policy composition (§7)
`anomaly_score_evolution.csv`	120 rows	Pooled mean Mahalanobis score per episode (§6.4)
`transition_entropy.csv`	30 rows	Per-episode $H(s' \mid s)$, single-agent checkpoint (§6.2)
`cluster_migration.csv`	12 rows	Regime dwell by phase, single-agent checkpoint (§6.3)

These artefacts feed directly into:

- **NB06 (Manifold Visualisation):** Q-tables and trajectory data for 3-D UMAP overlays of RL trajectories on the NB02 manifold, colour-coded by training phase and action.
- **NB07 (Final Experiment Analysis):** the trained-vs-random-baseline comparison methodology (§5) should be the template for any robustness sweep, not the original early-vs-late training-curve framing.
- **Paper §8:** key numbers are the regime-dwell reallocation deltas, the trained-vs-random-baseline comparison, and the $H(s' \mid s)$ reduction (-0.313 nats) — all independently reproduced. The policy corner composition (55.6%) is NB05's own result but did not independently reproduce (§7's caveat) and should not be cited in the paper without that context, or should be dropped in favour of the reproduced findings.

12 Next Steps

1. **NB06 — Manifold Visualisation.** Load Q-tables and trajectory data to produce publication-quality 3-D UMAP visualisations of RL trajectories overlaid on the NB02 manifold, coloured by training phase and action.
2. **NB07 — Final Robustness Analysis.** Include the RL layer in the planned multi-configuration robustness grid ($N \in \{5, 10, 20, 50\}$, $T \in \{500, 1000, 2000, 5000\}$, 10 seeds). Key question: does the trained (greedy) policy consistently beat the random-policy baseline (C1/C2's corrected framing) across all N/T configurations — *not* whether unstable fraction drops by some fixed relative percentage, which is no longer a meaningful target once the environment suppresses it near-floor by default.

3. **Extend the checkpoint methodology to the full pool.** §6's RQ1/RQ2/RQ4 results are currently a single-agent (agent_0000) case study. Repeating the controlled-epsilon checkpoint protocol across all 20 agents would let C4 and C6 be reported as pooled statistics with variance, rather than a single representative trajectory, and would clarify how much of §6.3's large stable-dwell shift generalises versus how much is agent_0000-specific.

4. **Resolve the `mid`-checkpoint confound.** Replace the zero-initialised Q-table used for the `early`/`mid` checkpoints with an explicit random or uniform tie-break policy, so that `mid`'s 60% greedy component is not deterministically `push_stable` by `argmax` convention.

5. **Paper §8.** NB05 constitutes the full quantitative evidence for the reinforcement learning section. The regime-*reallocation* framing (rather than unstable-*avoidance*) should be adopted throughout, and the environment causal-alignment fix (§3.2) is worth a brief methodological footnote given its generality to future RL-over- simulator work.

A Computational Environment

Component	Version / Note
Python	3.13.12
Environment manager	Nix flake dev shell (`flake.nix`)
NumPy, Pandas, SciPy	standard scientific stack
Matplotlib, Seaborn	standard scientific stack
scikit-learn	standard scientific stack
hmmlearn	HMM reference matrix from NB03
umap-learn	UMAP coordinates from NB02
Notebook execution	Jupyter (ipykernel), executed programmatically via `nbclient`

B File Manifest

All files below are under `outputs/figures/nb05/`.

File	Description
`fig_nb05_01_random_trajectory.png`	Random-policy verification episode
`fig_nb05_02_learning_curves.png`	Q-learning training curves (pool)
`fig_nb05_03_dwell_evolution.png`	Regime dwell evolution (pool)
`fig_nb05_04_manifold_alignment.png`	RL trajectory geometry, RQ1
`fig_nb05_05_transition_entropy.png`	Transition entropy reduction, RQ2
`fig_nb05_06_anomaly_dynamics.png`	Anomaly score evolution, RQ3
`fig_nb05_07_policy_analysis.png`	Learned policy analysis, C5
`fig_nb05_08_cluster_migration.png`	Cluster migration, RQ4

Generated by the LBSM research analysis pipeline; originally executed 2026-08-23, re-executed 2026-08-25 following the reward-key fix documented in §6.1. Dataset regenerated from `telemetry_n20_t2000.csv` with seed = 42. All numerical values sourced directly from cell outputs of `05_rl_behavioral_evolution.ipynb` following (a) a causal-alignment fix to `src/rl/environment.py`, (b) a correction of the evidence checklist's C1/C2 baseline (§3), and (c) a reward/`rl_reward` dict-key collision fix in the same file's `.trajectory` buffer construction (§6.1) — the only change between the two executions, confirmed by every table besides §6.1's reproducing to matching precision. This document is an internal technical record of Notebook 05 activities.